Program Requirements

Graduate Certificate in Underground Construction and Tunnel Engineering

The interdisciplinary Graduate Certificate in Underground Construction and Tunnel Engineering (UCTE) is comprised of the three signature courses listed below. The two anchor courses teach UCTE in hard rock and soft ground while the remaining course teaches underground infrastructure construction management principles.

Applicants for the certificate are required to be admitted into the combined BS + MS program or have an undergraduate degree in science or engineering, with geotechnical and mechanics of materials coursework, to be admitted into the certificate program. Students working toward the UCTE graduate certificate are required to successfully complete 10 credits, as detailed below. The courses taken for the graduate certificate can be used towards a master's or PhD degree at Mines.

CEEN523	UNDERGROUND CONSTRUCTION ENGINEERING IN SOFT GROUND	4.0
CEEN532	UNDERGROUND INFRASTRUCTURE CONSTRUCTION MANAGEMENT	3.0
or MNGN509	CONSTRUCTION ENGINEERING AND MANAGEMENT	
MNGN504	UNDERGROUND CONSTRUCTION ENGINEERING IN HARD ROCK	3.0
Total Semester Hrs		**10.0**

Program Director

Michael Mooney, UCTE Director, Grewcock Distinguished Chair

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Combined Undergraduate/ Graduate Degree Programs

OVERVIEW

Many degree programs offer Mines undergraduate students the opportunity to begin work on a professional master's degree, master's degree, or doctoral degree while completing the requirements for their bachelor's Degree. These combined bachelors–masters/doctoral programs have been created by Mines faculty in those situations where they have deemed it academically advantageous to treat undergraduate and graduate degree programs as a continuous and integrated process.

These are accelerated programs that can be valuable in fields of engineering and applied science where advanced education in technology and/or management provides the opportunity to be on a fast track for advancement to leadership positions. These programs can be valuable for students who want to get a head start on graduate education.

1. The combined programs at Mines offer several advantages to students who choose to enroll in them:
2. Students can earn a graduate degree in their undergraduate major or in a field that complements their undergraduate major.
3. Students who plan to go directly into industry leave Mines with additional specialized knowledge and skills which may allow them to enter their career path at a higher level and advance more rapidly. Alternatively, students planning on attending graduate school can get a head start on their graduate education.
4. Students can plan their undergraduate electives to satisfy prerequisites, thus ensuring adequate preparation for their graduate program.
5. Early assignment of graduate advisors permits students to plan optimum course selection and scheduling in order to complete their graduate program quickly.
6. Early acceptance into a combined degree program leading to a graduate degree assures students of automatic acceptance into full graduate status if they maintain good standing while in early-acceptance status.
7. In many cases, students will be able to complete both a bachelor's and a master's degrees in five years of total enrollment at Mines.

Students accepted into a combined program may substitute up to six credits of#graduate coursework (500-level) to fulfill requirements of both their undergraduate degree and their graduate degree.##These courses are referred to as double-counted courses. #If a course (required or elective) is cross listed as 4xx/5xx,#the 500-level version#must#be used for substitution purposes#to be#double counted.#Additionally, combined undergraduate students who register for a 500-level graduate course will by default have courses applied to the graduate record. The credits will be recorded on the graduate transcript and the credits will impact the graduate level GPA. Students should work closely with their advisor to select the courses for double counting by the add/drop date of the term in which the course is taken.

ENROLLING IN GRADUATE COURSES FOR A COMBINED PROGRAM

Once admitted into a graduate degree program, undergraduate students may enroll in 500-level courses.

Financial aid may be impacted. Email finaid@mines.edu for more information on financial aid impacts.

In addition, undergraduate seniors who have been granted admission through the combined degree program into thesis-based degree programs (master's or doctoral) may, with graduate advisor approval, register for 700-level research credits appropriate to their degree programs. With this single exception, while a combined degree program student is still completing their undergraduate degree, all of the conditions described in the undergraduate catalog for undergraduate enrollment in graduate-level courses apply. 700-level research credits are always applied to a student's graduate degree program.